Weekly Project Report – Week 4 Unified Generative AI Models via Command-Line Interface

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Objective

- Implement and train a Conditional Variational Autoencoder (CVAE) to generate and reconstruct digit images conditioned on class labels using the MNIST dataset.

Last Week

Last week, we implemented and trained a **Deep Convolutional GAN (DCGAN)** using PyTorch. The model consists of a convolutional **Generator** and **Discriminator**, designed to generate realistic images from random noise. We successfully trained the model on image data and visualized generated samples using fixed latent vectors.

Week-4 Completed

- Designed **encoder** and **decoder** networks with label embeddings.
- Applied **reparameterization trick** for latent sampling.
- Combined **reconstruction loss (BCE)** and **KL divergence** in the custom loss function.
- Trained model using the **Adam optimizer** with β weighting for KL loss.